LAB 6

# Code and Branch Coverage Analysis

Python application testing using pytest and Coverage.py

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|---|---|
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| **Application Under Test** | pizza_order.py |
| **Testing Tools** | Python 3.12.10, pytest 9.1.1, Coverage.py |
| **Platform** | Windows PowerShell |
| **Date** | 31 October 2026 |

Submitted as the Lab 6 practical report for Code and Branch Coverage Analysis.

# 1. Objective

The objective of this lab was to measure test completeness for a Python application, understand the difference between line (code) coverage and branch coverage, and design test cases that exercise normal conditions, boundary conditions, and error paths.

# 2. Application Overview

The application under test is **pizza_order.py**. It contains three functions:

| Function | Purpose and behaviour tested |
|---|---|
| pizza_price(size, toppings, is_cheese_burst) | Validates pizza size and toppings; calculates base price, topping charges, and cheese-burst surcharge; rejects invalid combinations. |
| order_discount(order_total, day, is_student) | Applies the minimum-order rule, Tuesday discount, weekend rule, regular weekday discount, and additional student discount. |
| delivery_status(minutes_taken, distance_km) | Determines whether an order is undeliverable, on time, late, or eligible for a free pizza. |

# 3. Tools and Environment

Testing was performed on Windows using Python 3.12.10. pytest was used to discover and execute the test functions, while Coverage.py collected statement and branch execution data. The application module was selected as the coverage source using `--source=pizza_order`.

# 4. Test Design and Execution

Separate test files were prepared for the two coverage exercises. The test cases cover valid pizza sizes, topping calculations, cheese-burst rules, invalid inputs, discount rules for different days and student status, and all delivery-status outcomes. Error conditions were checked with `pytest.raises()`.

### Code coverage command:

```
coverage run --source=pizza_order -m pytest -v
CodeCoverageTest_PES2UG24CS027_application.py
```

### Branch coverage command:

```
coverage run --branch --source=pizza_order -m pytest -v
BranchCoverageTest_PES2UG24CS027_application.py
```

### Report command:

```
coverage report -m
```

# Initial Test Run – Baseline Code Coverage

Before adding the expanded test cases, the original test file contained two test cases. Both tests passed, but the initial coverage report showed that only 14 of the 34 statements in **pizza_order.py** were executed.

```
PS D:\New folder (3)\Lab 6> pytest -v
============================================= test session starts =============================================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\ADHARSH\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: D:\New folder (3)\Lab 6
collected 2 items

test_pizza_order.py::test_plain_medium_pizza PASSED                                                  [ 50%]
test_pizza_order.py::test_on_time_delivery PASSED                                                    [100%]
                            coverage erase
>> coverage run --source=pizza_order -m pytest -v 2 passed in 0.04s ============================================
>> coverage report -mLab 6>
============================================= test session starts =============================================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\ADHARSH\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: D:\New folder (3)\Lab 6
collected 2 items

test_pizza_order.py::test_plain_medium_pizza PASSED                                                  [ 50%]
test_pizza_order.py::test_on_time_delivery PASSED                                                    [100%]

============================================== 2 passed in 0.07s ==============================================
Name             Stmts   Miss  Cover   Missing
------------------------------------------------
pizza_order.py      34     20    41%   6, 8, 11, 15-17, 22-32, 37, 40-42
------------------------------------------------
TOTAL               34     20    41%
```

*Figure 1. Initial terminal output: 2 tests passed; 20 of 34 statements were missed; baseline line coverage was 41%.*

This baseline result was used to identify untested parts of the application and guide the additional test cases in the following code coverage and branch coverage exercises.

# 5. Results – Code Coverage

The code coverage test suite completed successfully with 21 tests passing. Coverage.py reported that all 34 executable statements in pizza_order.py were executed.

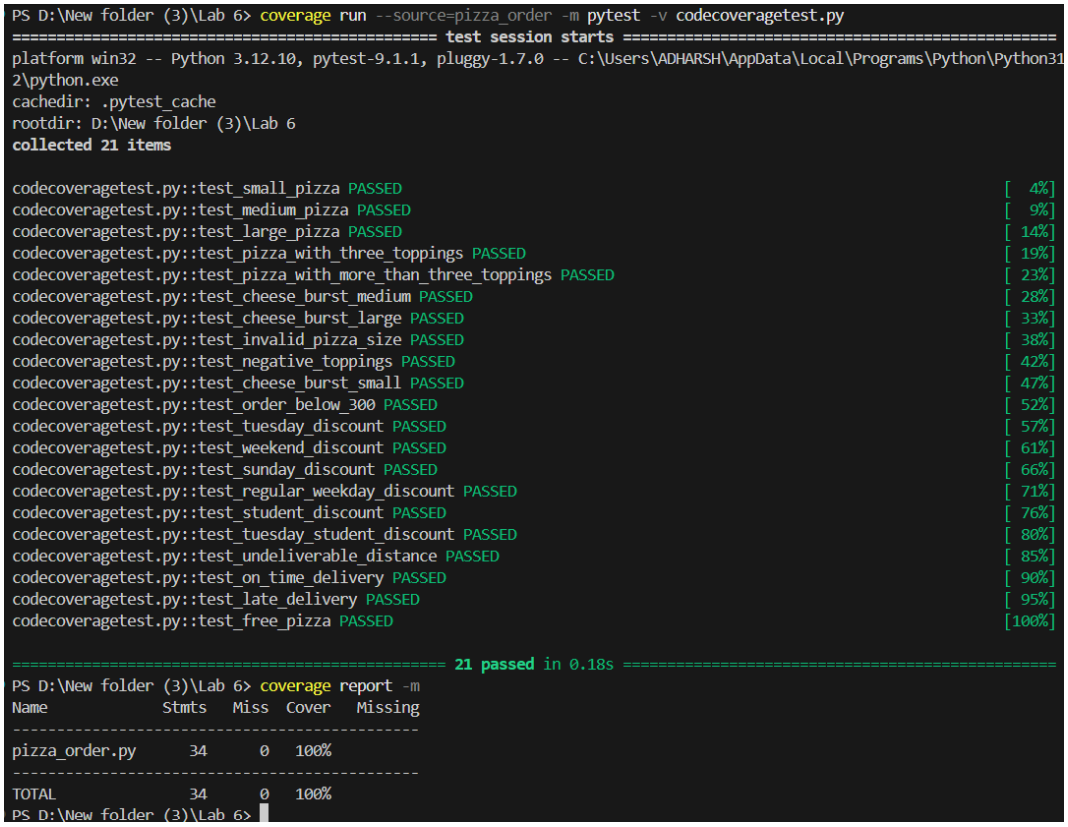



*Figure 1. Terminal output showing the passing tests and statement coverage report.*

| Measure | Observed result |
|---|---|
| Tests passed | 21 |
| Statements | 34 |
| Missed statements | 0 |
| Line coverage | 100% |

# 6. Results – Branch Coverage

The branch coverage test suite completed successfully with 21 tests passing. Coverage.py identified 24 branch destinations, with zero missing branches and zero partial branches.

```
PS D:\New folder (3)\Lab 6> coverage run --branch --source=pizza_order -m pytest -v BranchCoverageTest_PES2UG24CS027_application.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\ADHARSH\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
rootdir: D:\New folder (3)\Lab 6
collected 21 items

BranchCoverageTest_PES2UG24CS027_application.py::test_small_pizza PASSED                                                          [  4%]
BranchCoverageTest_PES2UG24CS027_application.py::test_medium_pizza PASSED                                                         [  9%]
BranchCoverageTest_PES2UG24CS027_application.py::test_large_pizza PASSED                                                          [ 14%]
BranchCoverageTest_PES2UG24CS027_application.py::test_pizza_with_three_toppings PASSED                                            [ 19%]
BranchCoverageTest_PES2UG24CS027_application.py::test_pizza_with_more_than_three_toppings PASSED                                  [ 23%]
BranchCoverageTest_PES2UG24CS027_application.py::test_cheese_burst_medium PASSED                                                  [ 28%]
BranchCoverageTest_PES2UG24CS027_application.py::test_cheese_burst_large PASSED                                                   [ 33%]
BranchCoverageTest_PES2UG24CS027_application.py::test_invalid_pizza_size PASSED                                                   [ 38%]
BranchCoverageTest_PES2UG24CS027_application.py::test_negative_toppings PASSED                                                    [ 42%]
BranchCoverageTest_PES2UG24CS027_application.py::test_cheese_burst_small PASSED                                                   [ 47%]
BranchCoverageTest_PES2UG24CS027_application.py::test_order_below_300 PASSED                                                      [ 52%]
BranchCoverageTest_PES2UG24CS027_application.py::test_tuesday_discount PASSED                                                     [ 57%]
BranchCoverageTest_PES2UG24CS027_application.py::test_weekend_discount PASSED                                                     [ 61%]
BranchCoverageTest_PES2UG24CS027_application.py::test_sunday_discount PASSED                                                      [ 66%]
BranchCoverageTest_PES2UG24CS027_application.py::test_regular_weekday_discount PASSED                                             [ 71%]
BranchCoverageTest_PES2UG24CS027_application.py::test_student_discount PASSED                                                     [ 76%]
BranchCoverageTest_PES2UG24CS027_application.py::test_tuesday_student_discount PASSED                                             [ 80%]
BranchCoverageTest_PES2UG24CS027_application.py::test_undeliverable_distance PASSED                                               [ 85%]
BranchCoverageTest_PES2UG24CS027_application.py::test_on_time_delivery PASSED                                                     [ 90%]
BranchCoverageTest_PES2UG24CS027_application.py::test_late_delivery PASSED                                                        [ 95%]
BranchCoverageTest_PES2UG24CS027_application.py::test_free_pizza PASSED                                                           [100%]

============================================================ 21 passed in 0.18s =============================================================
PS D:\New folder (3)\Lab 6> coverage report -m
Name            Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------
pizza_order.py      34      0     24      0   100%
-----------------------------------------------------------
TOTAL               34      0     24      0   100%
PS D:\New folder (3)\Lab 6>
```

*Figure 2. Terminal output showing the passing tests and branch coverage report.*

| Measure | Observed result |
|---|---|
| Tests passed | 21 |
| Statements | 34 |
| Missed statements | 0 |
| Branches | 24 |
| Missing / partial branches | 0 |
| Coverage | 100% |

# 7. Analysis: Line Coverage vs. Branch Coverage

**Line coverage** measures whether executable source statements were run at least once. The result was 34 out of 34 statements executed.

**Branch coverage** checks whether the possible outcomes of decision points were exercised. The report showed 24 branches and zero missing or partial branches.

The two metrics provide different information: executing every statement does not, by itself, prove that every True and False decision outcome has been tested. Running the branch-enabled coverage command provided evidence for the decision paths as well.

# 8. GitHub Repository and Version Control

The Lab 6 files are available in the following repository folder:

PES2UG24CS027_SE_LAB – Lab 6 folder

| Coverage stage | Commit message | Commit link |
|---|---|---|
| Code coverage | Add code coverage tests with 100 percent coverage | 2204ed2 – View commit |
| Branch coverage | Add branch coverage tests with 100 percent coverage | 3137396 – View commit |

The code coverage commit contains the code coverage test file and its terminal screenshot. The branch coverage commit contains the branch coverage test file and its terminal screenshot. Both commits were checked in the specified GitHub repository.

# 9. Conclusion

The Lab 6 testing objectives were achieved for the pizza ordering application. The final test runs completed with 21 passing tests in each test suite. The statement coverage report showed complete line coverage, and the branch coverage report showed all 24 branches exercised. Separate test files and terminal screenshots were prepared, and the work was organized into separate Git commits as required.

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*End of Report*